Bagging Predictors

Leo Breiman, *Machine Learning* 1996

Sohan Shingade · DSC 180A · Week 3

1. The task: predict diabetes from 8 numbers

Patient	Glucose	BMI	Age	…	Diabetic?
A	148	33.6	50		yes
B	85	26.6	31		no
C	183	23.3	32		yes
D	89	28.1	21		no
E	137	43.1	33		yes
F	116	25.6	30		no

768 patients like this (Pima Indians Diabetes, Kaggle). Breiman used this same dataset.

2. One decision tree is a shaky guess

Tree trained on A–F

```
glucose > 127 ?
  yes → diabetic
  no  → not diabetic
```

Swap out two patients, retrain

```
BMI > 30 ?
  yes → age > 40 ? …
  no  → not diabetic
```

Change a little data and the whole tree changes. Breiman calls this **unstable**.

The single tree in the paper got 23% of diabetes patients wrong.

3. Noisy guesses average out, but I only have one dataset

If I had 50 **independent** datasets, I could train 50 trees and let them vote. Their individual mistakes would cancel.

one dataset → one tree → one noisy guess

I can't go collect 50 more sets of 768 patients.

So: can I fake 50 datasets from the one I have?

4. Bootstrap: resample your own data, with replacement

Draw 6 patients from {A, B, C, D, E, F} at random. Put each one back before the next draw.

Copy	Draws	Repeated	Left out
1	A A C D E E	A, E	B, F
2	B B C C D F	B, C	A, E
3	A B D E F F	F	C

Each copy is the same size as the original, but some rows repeat and some are missing.

"Bootstrap": Efron, 1979. Breiman's idea was to feed these copies to a learner.

5. Why a resample is a fair fake of a new dataset

What actually happened

The world has millions of people.
My 768 patients are a **random sample** of them.
Another study would get a *different* 768.

What bootstrap imitates

Treat my 768 as if they *were* the world.
A bootstrap copy is a **random sample** of them.
Another copy is a *different* 768.

The amount a tree **wobbles** between bootstrap copies is close to how much it would wobble between real studies. That is the whole bet.

It works because the 768 are a decent stand-in for the population. With 6 patients it would not.

6. Each copy sees about 63% of the patients

Chance a given patient is **never** drawn in 768 draws: $(1 - 1/768)^{768} \approx 1/e \approx$ **37%**.

So each bootstrap copy contains about **63%** of the distinct patients, with the rest filled by repeats.

different 63% → different tree

That is exactly what we want: 50 copies give 50 genuinely different trees.

7. Bagging = bootstrap + aggregate

```
for b in 1 … 50:
    copy_b = bootstrap sample of the data
    tree_b = train a tree on copy_b

to predict a new patient:
    ask all 50 trees
    label  →  majority vote
    number →  average
```

Nothing inside the tree learner changes. Breiman: all you add is "a loop in front … and a back end that does the aggregation."

8. Three trees vote on a new patient

New patient: glucose 130, BMI 35, age 45.

Tree	Trained on	Its first question	Says
1	copy 1	glucose > 127?	diabetic
2	copy 2	BMI > 30?	diabetic
3	copy 3	age > 48?	not diabetic

$$2 - 1 \rightarrow \text{diabetic}$$

A 2–1 vote also says something a single tree never can: **this one isn't certain.**

9. Why it works: averaging cancels noise

Unstable learner (tree)

50 copies → 50 different trees
Averaging their answers removes the wobble

big gain

Stable learner (nearest neighbour)

50 copies → 50 nearly identical answers
Nothing to average away

no gain

Breiman's proof: the squared error of the average is never above the average squared error. The gap is the learner's variance. No variance, no gap.

> *"Bagging unstable classifiers usually improves them. Bagging stable classifiers is not a good idea."*

10. Breiman's results: better on every dataset

Dataset	One tree	50 bagged	Decrease
waveform	29.0	19.4	33%
heart (UCSD Medical Center)	10.0	5.3	47%
breast cancer	6.0	4.2	30%
ionosphere	11.2	8.6	23%
diabetes	**23.4**	**18.8**	**20%**
glass	32.0	24.9	22%
soybean	14.5	10.6	27%

Test error %, averaged over 100 random splits. Nearest neighbour: **no change** on any dataset.

How many copies? 10 gets most of the gain, 25 nearly all. *"More than 25 is love's labor lost."*

11. My run: same story on the diabetes data

50 random 80/20 splits, sklearn trees. Error is mean ± 95% CI.

Method	Test error %	Train (ms)	Predict (μs / patient)
one tree	30.1 ± 1.0	1.1	0.2
50 bagged trees	**23.9 ± 0.8**	45	9.6
nearest neighbour	29.5 ± 0.8	0.2	2.9
50 bagged NN	29.4 ± 0.8	18	106

- **Accuracy:** 21% less error (Breiman: 20%). Bagging won on 49 of 50 splits.
- **Confidence:** when the 50-tree vote is lopsided (>45–5) it is right **92%** of the time; when it is close (<30–20) only **57%**.
- **Speed:** ~40× slower to train and predict, but still 10 μs per patient.

12. Takeaways

1. **Bootstrap** fakes new datasets by resampling the one you have. It is a fair fake because your sample stands in for the population.
2. **Bagging** = bootstrap copies + vote. The learner is untouched.
3. It fixes **unstable** learners (trees: –20% to –47%). Stable ones gain nothing.
4. The **vote split** is a free confidence estimate.
5. Each copy trains with **zero communication** to the others. Breiman: "almost a dream procedure for parallel computing." That is why it maps onto a swarm of learner agents and one coordinator that votes.

Code: `slides/bagging_1996/bagging_pima_diabetes.ipynb`